

# NanoBanana Study Report: polyp | stage0 | nanobanana\_polyp\_s

run\_dir: D:\Projects\gemini\_image\_segmentation\results\_nanobanana\polyp\gemini-3.1-flash-image-pr

run\_id: nanobanana\_polyp\_smoke\_stage0\_20260302-012659

stage: stage0

dataset: polyp

target: polyp

model: gemini-3.1-flash-image-preview

n\_tasks: 40

n\_rows: 40

n\_failures: 0

workers: 6

Interpretation notes:

- PRIMARY excludes duplicate/mask-source flagged retrieval runs per config.
- SENSITIVITY includes all audited rows.
- Delta tables compare each tool mode against CLOSED on paired rows.

## Stall/Long-Running Events

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```

## Stall/Long-Running Events (cont.)

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{"elapsed_seconds": 273.234, "task": "image=01dcc13a-85ab-4d6e-b42b-5d631e076ac2.jpg mode=text_in  
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```

Summary By Tool Mode (PRIMARY)

| tool_mode  | n       | mean_iou | mean_dice | mean_precision | mean_recall | qc_pass_rate | iou_ci_lower | iou_ci_upper | dice_ci_lower | dice_ci_upper |
|------------|---------|----------|-----------|----------------|-------------|--------------|--------------|--------------|---------------|---------------|
| closed     | 10.0000 | 0.8039   | 0.8765    | 0.9703         | 0.8304      | 0.0000       | 0.6706       | 0.9075       | 0.7761        | 0.9462        |
| image      | 10.0000 | 0.7043   | 0.7851    | 0.8011         | 0.8915      | 0.0000       | 0.5175       | 0.8644       | 0.6119        | 0.9204        |
| text       | 10.0000 | 0.7296   | 0.8233    | 0.9057         | 0.8197      | 0.0000       | 0.5903       | 0.8603       | 0.7178        | 0.9187        |
| text_image | 10.0000 | 0.7502   | 0.8209    | 0.8200         | 0.9273      | 0.0000       | 0.5750       | 0.8953       | 0.6425        | 0.9415        |

Summary By Tool Mode (SENSITIVITY)

| tool_mode  | n       | mean_iou | mean_dice | mean_precision | mean_recall | qc_pass_rate | iou_ci_lower | iou_ci_upper | dice_ci_lower | dice_ci_upper |
|------------|---------|----------|-----------|----------------|-------------|--------------|--------------|--------------|---------------|---------------|
| closed     | 10.0000 | 0.8039   | 0.8765    | 0.9703         | 0.8304      | 0.0000       | 0.6704       | 0.8966       | 0.7775        | 0.9484        |
| image      | 10.0000 | 0.7043   | 0.7851    | 0.8011         | 0.8915      | 0.0000       | 0.5152       | 0.8715       | 0.6172        | 0.9264        |
| text       | 10.0000 | 0.7296   | 0.8233    | 0.9057         | 0.8197      | 0.0000       | 0.5952       | 0.8557       | 0.7199        | 0.9238        |
| text_image | 10.0000 | 0.7502   | 0.8209    | 0.8200         | 0.9273      | 0.0000       | 0.5725       | 0.8973       | 0.6621        | 0.9387        |

Delta Vs CLOSED (PRIMARY)

| tool_mode  | n       | mean_delta_iou | mean_delta_dice |
|------------|---------|----------------|-----------------|
| image      | 10.0000 | -0.0996        | -0.0914         |
| text       | 10.0000 | -0.0743        | -0.0532         |
| text_image | 10.0000 | -0.0538        | -0.0556         |

QC Failure Summary (PRIMARY)

| tool_mode  | qc_fail_count | total   | qc_fail_rate |
|------------|---------------|---------|--------------|
| closed     | 10.0000       | 10.0000 | 1.0000       |
| image      | 10.0000       | 10.0000 | 1.0000       |
| text       | 10.0000       | 10.0000 | 1.0000       |
| text_image | 10.0000       | 10.0000 | 1.0000       |

# Retrieval Audit Summary

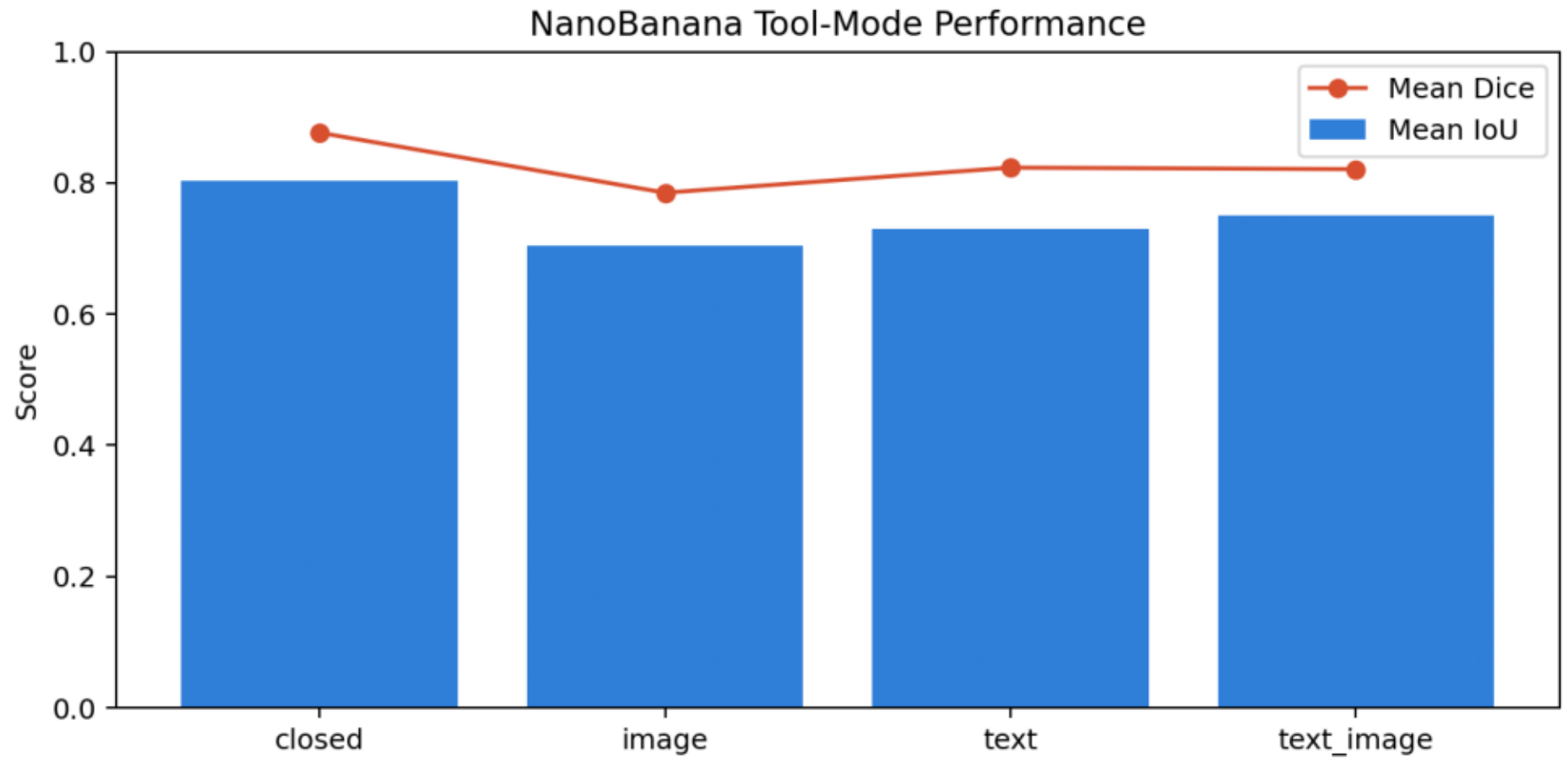
| tool_mode  | duplicate_rate | mask_source_rate | audit_unavailable_rate | audit_unavailable_count | n       |
|------------|----------------|------------------|------------------------|-------------------------|---------|
| closed     | 0.0000         | 0.0000           | 1.0000                 | 10.0000                 | 10.0000 |
| image      | 0.0000         | 0.0000           | 1.0000                 | 10.0000                 | 10.0000 |
| text       | 0.0000         | 0.0000           | 1.0000                 | 10.0000                 | 10.0000 |
| text_image | 0.0000         | 0.0000           | 1.0000                 | 10.0000                 | 10.0000 |



# Analysis Partition Counts

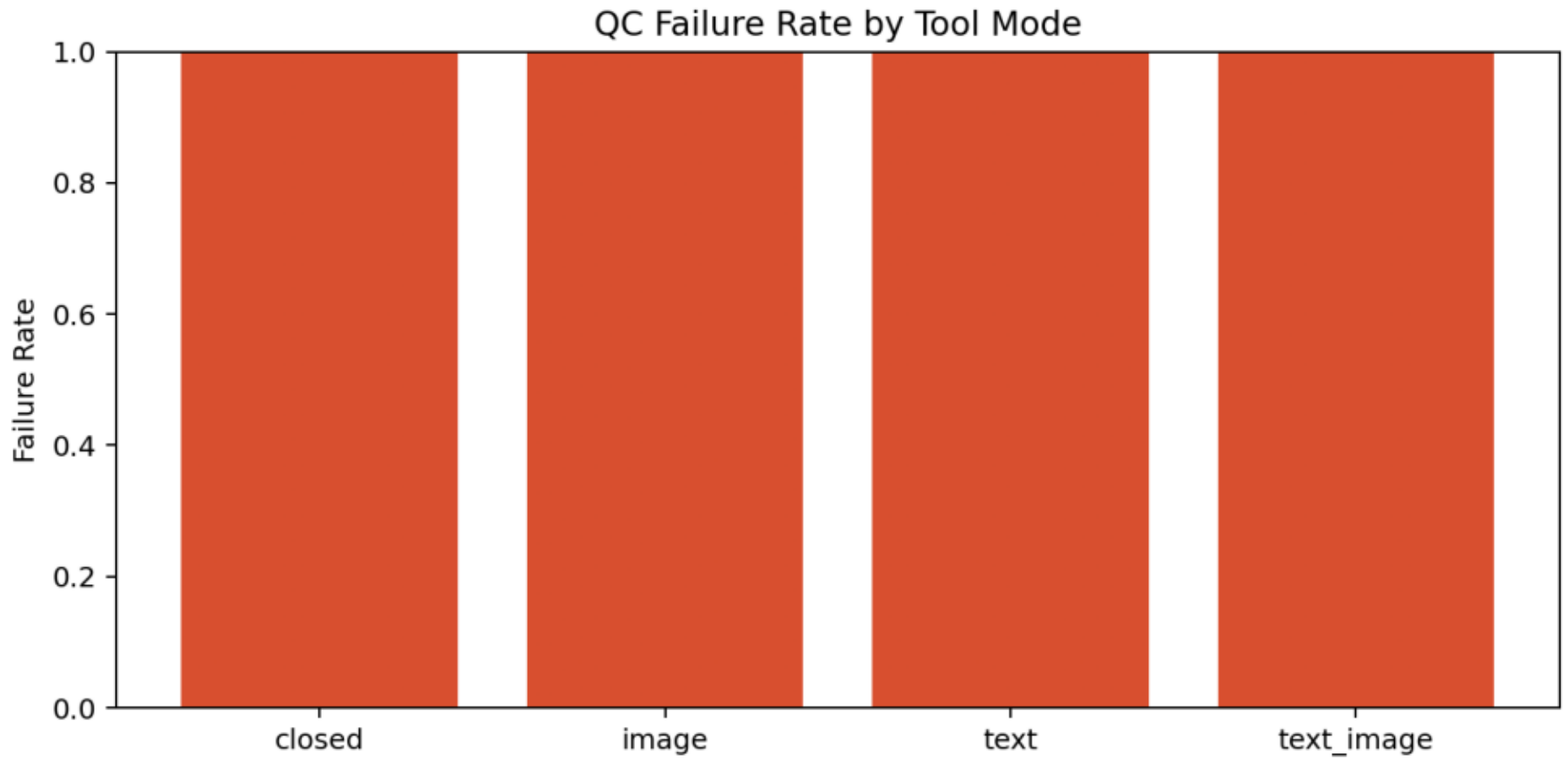
| tool_mode  | n_total | n_primary | n_sensitivity | n_audit_unavailable |
|------------|---------|-----------|---------------|---------------------|
| closed     | 10.0000 | 10.0000   | 10.0000       | 10.0000             |
| image      | 10.0000 | 10.0000   | 10.0000       | 10.0000             |
| text       | 10.0000 | 10.0000   | 10.0000       | 10.0000             |
| text_image | 10.0000 | 10.0000   | 10.0000       | 10.0000             |

# Primary Performance Plot



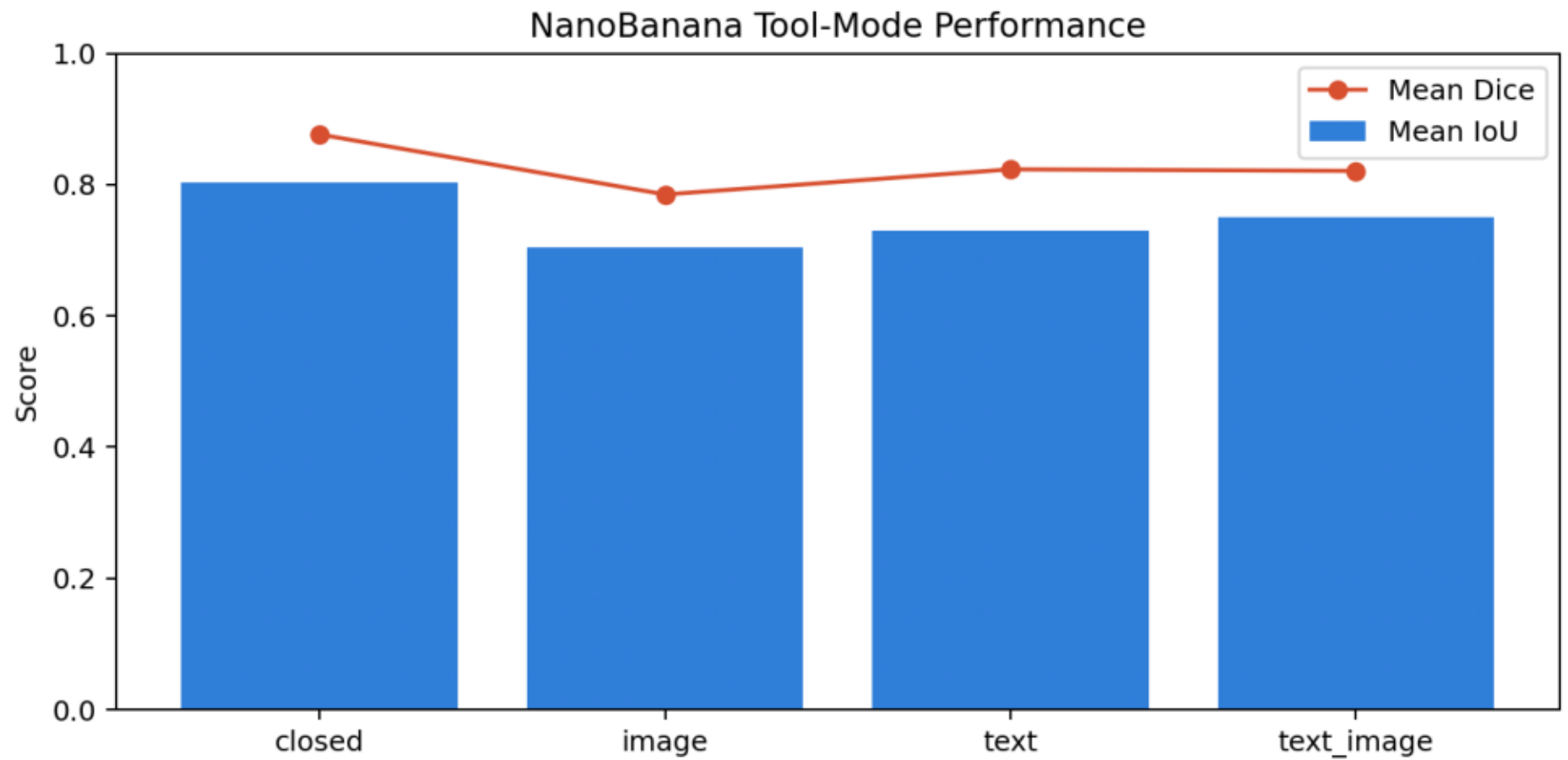
Mean IoU bars with Mean Dice line by tool mode.

# Primary QC Failure Plot



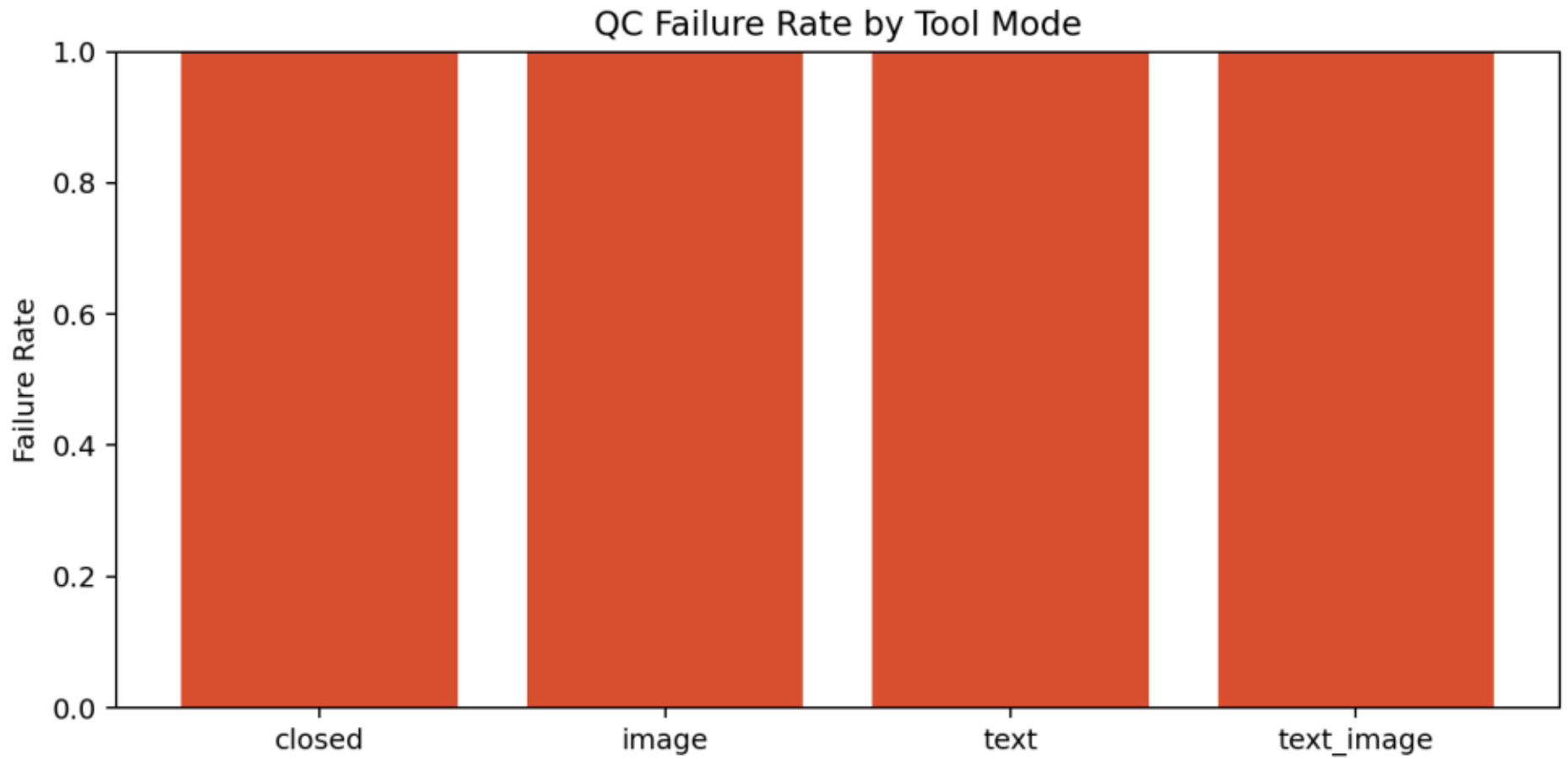
QC failure rate by tool mode.

# Sensitivity Performance Plot



Sensitivity-set mean IoU/Dice by tool mode.

## Sensitivity QC Failure Plot



Sensitivity-set QC failure rate by tool mode.

Per-Run Results (Excerpt)

| image_name    | tool_mode  | iou    | dice   | precision | recall | qc_pass | selected_atte | retrieval_dup | retrieval_mas | analysis_primar |
|---------------|------------|--------|--------|-----------|--------|---------|---------------|---------------|---------------|-----------------|
| 0004a718-546c | closed     | 0.8486 | 0.9181 | 0.8713    | 0.9701 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0004a718-546c | text       | 0.4243 | 0.5958 | 0.4261    | 0.9901 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0004a718-546c | image      | 0.1856 | 0.3131 | 0.1890    | 0.9102 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0004a718-546c | text_image | 0.4519 | 0.6225 | 0.4546    | 0.9868 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0017b7c7-90f8 | closed     | 0.6581 | 0.7938 | 0.9875    | 0.6637 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0017b7c7-90f8 | text       | 0.6541 | 0.7909 | 0.9873    | 0.6597 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0017b7c7-90f8 | image      | 0.5783 | 0.7328 | 0.9850    | 0.5834 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0017b7c7-90f8 | text_image | 0.1043 | 0.1890 | 0.1043    | 1.0000 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0048d8c5-b59c | closed     | 0.9551 | 0.9770 | 0.9916    | 0.9629 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0048d8c5-b59c | text       | 0.9520 | 0.9754 | 0.9938    | 0.9577 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0048d8c5-b59c | image      | 0.9527 | 0.9758 | 0.9931    | 0.9590 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0048d8c5-b59c | text_image | 0.9519 | 0.9754 | 0.9929    | 0.9584 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 00f98835-8fd8 | closed     | 0.8696 | 0.9303 | 0.9998    | 0.8698 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 00f98835-8fd8 | text       | 0.8721 | 0.9317 | 0.9996    | 0.8725 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 00f98835-8fd8 | image      | 0.8692 | 0.9300 | 0.9997    | 0.8694 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 00f98835-8fd8 | text_image | 0.8698 | 0.9304 | 0.9994    | 0.8702 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01503109-d81f | closed     | 0.8629 | 0.9264 | 0.8952    | 0.9599 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01503109-d81f | text       | 0.6774 | 0.8077 | 0.6898    | 0.9742 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01503109-d81f | image      | 0.6866 | 0.8142 | 0.6999    | 0.9731 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01503109-d81f | text_image | 0.6849 | 0.8130 | 0.6963    | 0.9766 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0198bc50-169e | closed     | 0.9342 | 0.9660 | 0.9887    | 0.9443 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0198bc50-169e | text       | 0.9367 | 0.9673 | 0.9898    | 0.9458 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0198bc50-169e | image      | 0.9372 | 0.9676 | 0.9892    | 0.9469 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 0198bc50-169e | text_image | 0.9317 | 0.9646 | 0.9886    | 0.9418 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01c95ee8-79cc | closed     | 0.9287 | 0.9630 | 0.9888    | 0.9385 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01c95ee8-79cc | text       | 0.9286 | 0.9630 | 0.9938    | 0.9340 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01c95ee8-79cc | image      | 0.9312 | 0.9644 | 0.9955    | 0.9351 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01c95ee8-79cc | text_image | 0.9305 | 0.9640 | 0.9957    | 0.9342 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |

Per-Run Results (Excerpt) (cont. 2)

| image_name     | tool_mode  | iou    | dice   | precision | recall | qc_pass | selected_atte | retrieval_dup | retrieval_mas | analysis_primar |
|----------------|------------|--------|--------|-----------|--------|---------|---------------|---------------|---------------|-----------------|
| 01dbf397-c0fc- | closed     | 0.9569 | 0.9780 | 0.9909    | 0.9654 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01dbf397-c0fc- | text       | 0.9622 | 0.9808 | 0.9904    | 0.9713 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01dbf397-c0fc- | image      | 0.1837 | 0.3103 | 0.1837    | 1.0000 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01dbf397-c0fc- | text_image | 0.9632 | 0.9812 | 0.9885    | 0.9741 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01dcc13a-85ab- | closed     | 0.7037 | 0.8261 | 0.9934    | 0.7069 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01dcc13a-85ab- | text       | 0.5309 | 0.6935 | 0.9920    | 0.5332 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01dcc13a-85ab- | image      | 0.7596 | 0.8633 | 0.9907    | 0.7650 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01dcc13a-85ab- | text_image | 0.6496 | 0.7876 | 0.9937    | 0.6523 | 0.0000  | 1.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01e6cfd7-bdb6- | closed     | 0.3216 | 0.4866 | 0.9953    | 0.3220 | 0.0000  | 3.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01e6cfd7-bdb6- | text       | 0.3579 | 0.5271 | 0.9947    | 0.3586 | 0.0000  | 3.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01e6cfd7-bdb6- | image      | 0.9595 | 0.9793 | 0.9856    | 0.9731 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |
| 01e6cfd7-bdb6- | text_image | 0.9641 | 0.9817 | 0.9854    | 0.9781 | 0.0000  | 2.0000        | 0.0000        | 0.0000        | 1.0000          |